ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 53	$x^2+y^2=(x+y)^2-2xy$
2	**(a)** 49	$x^2+y^2=(x-y)^2+2xy$
3	**(d)** 24000	$(512+488)(512-488)$
4	**(b)** 7	Square both sides
5	**(c)** 25	$p+q=a,\ pq=b$
6	**(d)** 141	25^2-22^2
7	**(b)** 47	$n^2-(n-1)^2=2n-1$
8	**(b)** $ab=0$	Compare with $a^2+2ab+b^2$
9	**(a)** Both A and R are true and R is the correct explanation of A.	Put $a=x,\ b=5$ in the identity: $x^2+2(x)(5)+25$.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$(a-b)^2=a^2-2ab+b^2=(b-a)^2$ is true. R is also true but it is a different identity and does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2+y^2=(x+y)^2-2xy$ (1) $=144-70=74$ (1).	1+1
12	$(612+388)(612-388)=1000\times224$ (1) $=224000$ (1).	1+1
13	$p+q=8,\ pq=15$ (1); $p^2+q^2=8^2-2(15)=34$ (1).	1+1
14	$x^2+y^2=(x+y)^2-2xy$ (1) $=64-30=34$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2+y^2=(x-y)^2+2xy=16+42=58$ (1.5); $(x+y)^2=(x-y)^2+4xy=16+84=100$ (1.5).	1½+1½
16	$(3x+2)^2=9x^2+12x+4$ (1); $(3x-2)^2=9x^2-12x+4$ (1); difference $=24x$ (1).	1+1+1
17	$a^2+b^2=(10)^2-2(24)=52$ (1.5); $(a-b)^2=a^2+b^2-2ab=52-48=4$ (1.5).	1½+1½
	OR: $x^2+y^2=(x-y)^2+2xy=36+80=116$ (1.5); $(x+y)^2=(x-y)^2+4xy=36+160=196$ (1.5).	1½+1½
18	$(4x+5)^2=16x^2+40x+25$ (1); $(4x-5)^2=16x^2-40x+25$ (1); difference $=80x$ (1).	1+1+1
	OR: $a^2+b^2=(13)^2-2(40)=89$ (1.5); $(a-b)^2=a^2+b^2-2ab=89-80=9$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(x+5)(x+2)=x^2+7x+10$ (1). (ii) $15\times12=180$ m² (1). (iii) $2(2x+7)=4x+14$ m (1). (iv) Bed area $(x+3)^2=x^2+6x+9$; difference $=x+1$ (1); for $x=10$: $180-169=11=x+1$ (1).	1+1+1+2
20	Area $=(4x+3)(4x-3)=16x^2-9$ (2); at $x=6$: 567 m² (1); square $=16x^2$, difference $=9$ m² (2).	2+1+2
	OR: $xy=\dfrac{(16)^2-146}{2}=55$ (2); $(x-y)^2=146-2(55)=36$ (2); $x-y=6$ (1).	2+2+1

Mistakes to avoid:

- Writing $(a+b)^2=a^2+b^2$; the middle term $2ab$ is missed.
- Writing $(a-b)^2=a^2-b^2$; correct is $a^2-2ab+b^2$, while $a^2-b^2=(a+b)(a-b)$.
- Multiplying only the first term of a bracket: $3(x+4)=3x+4$ instead of $3x+12$.
- Sign errors when multiplying: $-2\times(-3x)=+6x$ and $(x-3)(x-2)$ has $+6$ as the constant term.
- Adding unlike terms: $2x+3x^2=5x^3$; x and x^2 are not like terms.
- Squaring a product without squaring the coefficient: $(3x)^2=9x^2$, not $3x^2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 508000	$(754+246)(754-246)$
2	**(c)** 14	Square both sides
3	**(a)** 41	$p+q=a$, $pq=b$
4	**(b)** 511	40^2-33^2
5	**(b)** 69	$n^2-(n-1)^2=2n-1$
6	**(c)** a^2+b^2	Expand
7	**(a)** 97	$x^2+y^2=(x+y)^2-2xy$
8	**(c)** 73	$x^2+y^2=(x-y)^2+2xy$
9	**(c)** A is true but R is false.	A is the identity $(a+b)(a-b)=a^2-b^2$; R is false because $(a+b)^2=a^2+2ab+b^2$.
10	**(d)** A is false but R is true.	$(2x+3)(x+1)=2x^2+2x+3x+3=2x^2+5x+3$, so A is false; R is the correct rule.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(845+155)(845-155)=1000\times690$ (1) $=690000$ (1).	1+1
12	$p+q=12$, $pq=32$ (1); $p^2+q^2=12^2-2(32)=80$ (1).	1+1
13	$x^2+y^2=(x+y)^2-2xy$ (1) $=196-66=130$ (1).	1+1
14	$(703+297)(703-297)=1000\times406$ (1) $=406000$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2+y^2=(x-y)^2+2xy=9+56=65$ (1.5); $(x+y)^2=(x-y)^2+4xy=9+112=121$ (1.5).	1½+1½
16	$(5x+3)^2=25x^2+30x+9$ (1); $(5x-3)^2=25x^2-30x+9$ (1); difference $=60x$ (1).	1+1+1
17	$a^2+b^2=(9)^2-2(18)=45$ (1.5); $(a-b)^2=a^2+b^2-2ab=45-36=9$ (1.5).	1½+1½
	OR: $x^2+y^2=(x-y)^2+2xy=25+72=97$ (1.5); $(x+y)^2=(x-y)^2+4xy=25+144=169$ (1.5).	1½+1½
18	$(2x+7)^2=4x^2+28x+49$ (1); $(2x-7)^2=4x^2-28x+49$ (1); difference $=56x$ (1).	1+1+1
	OR: $a^2+b^2=(15)^2-2(56)=113$ (1.5); $(a-b)^2=a^2+b^2-2ab=113-112=1$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $98=100-2$ (1). (ii) $(100-2)^2=10000-400+4=9604$ (1). (iii) $(100+3)(100-3)=10000-9=9991$ (1). (iv) $(52+48)(52-48)=100\times4=400$ (1); identity $a^2-b^2=(a+b)(a-b)$ (1).	1+1+1+2
20	Area $=(2x+7)(2x-7)=4x^2-49$ (2); at $x=8$: 207 m^2 (1); square $=4x^2$, difference $=49$ m^2 (2).	2+1+2
	OR: $xy=\dfrac{(19)^2-193}{2}=84$ (2); $(x-y)^2=193-2(84)=25$ (2); $x-y=5$ (1).	2+2+1

Mistakes to avoid:

- Writing $(a+b)^2=a^2+b^2$; the middle term $2ab$ is missed.
- Writing $(a-b)^2=a^2-b^2$; correct is $a^2-2ab+b^2$, while $a^2-b^2=(a+b)(a-b)$.
- Multiplying only the first term of a bracket: $3(x+4)=3x+4$ instead of $3x+12$.
- Sign errors when multiplying: $-2\times(-3x)=+6x$ and $(x-3)(x-2)$ has $+6$ as the constant term.
- Adding unlike terms: $2x+3x^2=5x^3$; x and x^2 are not like terms.
- Squaring a product without squaring the coefficient: $(3x)^2=9x^2$, not $3x^2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 65	$p+q=a, pq=b$
2	**(d)** 272	36^2-32^2
3	**(c)** 95	$n^2-(n-1)^2=2n-1$
4	**(a)** $a=b$ or $a=-b$	$(a+b)(a-b)=0$
5	**(a)** 58	$x^2+y^2=(x+y)^2-2xy$
6	**(a)** 74	$x^2+y^2=(x-y)^2+2xy$
7	**(c)** 806000	$(903+97)(903-97)$
8	**(d)** 34	Square both sides
9	**(a)** Both A and R are true and R is the correct explanation of A.	$49\times51=(50-1)(50+1)=50^2-1^2=2500-1=2499.$
10	**(c)** A is true but R is false.	$(3x)^2=3x\times3x=9x^2$ so A is true; R is false because the coefficient is also squared.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p+q=15, pq=54$ (1); $p^2+q^2=15^2-2(54)=117$ (1).	1+1
12	$x^2+y^2=(x+y)^2-2xy$ (1) $=289-120=169$ (1).	1+1
13	$(960+40)(960-40)=1000\times920$ (1) $=920000$ (1).	1+1
14	$p+q=9, pq=14$ (1); $p^2+q^2=9^2-2(14)=53$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x^2+y^2=(x-y)^2+2xy=49+60=109$ (1.5); $(x+y)^2=(x-y)^2+4xy=49+120=169$ (1.5).	1½+1½
16	$(6x+1)^2=36x^2+12x+1$ (1); $(6x-1)^2=36x^2-12x+1$ (1); difference $=24x$ (1).	1+1+1
17	$a^2+b^2=(11)^2-2(30)=61$ (1.5); $(a-b)^2=a^2+b^2-2ab=61-60=1$ (1.5).	1½+1½
	OR: $x^2+y^2=(x-y)^2+2xy=64+96=160$ (1.5); $(x+y)^2=(x-y)^2+4xy=64+192=256$ (1.5).	1½+1½
18	$(3x+8)^2=9x^2+48x+64$ (1); $(3x-8)^2=9x^2-48x+64$ (1); difference $=96x$ (1).	1+1+1
	OR: $a^2+b^2=(14)^2-2(45)=106$ (1.5); $(a-b)^2=a^2+b^2-2ab=106-90=16$ (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 6 (1). (ii) $6(2a+3b)$ (1). (iii) $6(30+30)=360$; check $180+180=360$ (1). (iv) $4(x^2-9)=4(x+3)(x-3)$ (1); at $x=5$: $4\times8\times2=64$ and $4(25)-36=64$ (1).	1+1+1+2
20	Area $=(7x+2)(7x-2)=49x^2-4$ (2); at $x=4$: 780 m^2 (1); square $=49x^2$, difference $=4$ m^2 (2).	2+1+2
	OR: $xy=\dfrac{(15)^2-173}{2}=26$ (2); $(x-y)^2=173-2(26)=121$ (2); $x-y=11$ (1).	2+2+1

Mistakes to avoid:

- Writing $(a+b)^2=a^2+b^2$; the middle term $2ab$ is missed.
- Writing $(a-b)^2=a^2-b^2$; correct is $a^2-2ab+b^2$, while $a^2-b^2=(a+b)(a-b)$.
- Multiplying only the first term of a bracket: $3(x+4)=3x+4$ instead of $3x+12$.
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